

Multi-Omics Muscle Atrophy Countermeasure Pipeline

Consensus Catabolic Signature, LINCS Drug Screening, and Vegan Nutritional Translation

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Executive Summary

This report presents a reproducible multi-omics pipeline that integrates 22 NASA OSDR datasets (rodent spaceflight, human astronaut, tissue chips, ground analogs) with GEO and ALSDA physiological data to derive a consensus cross-species muscle atrophy/catabolic signature. A random-effects meta-analysis across 10 transcriptomic studies (32,770 genes) identified 141 differentially expressed genes (77 up-regulated, 64 down-regulated; $|\log_2FC| > 0.25$, $p_{adj} < 0.05$). The signature was characterized by upregulation of mitochondrial electron transport genes (MT-ND1, MT-CO1, MT-ATP6) and the atrophy ubiquitin ligase FBXO32, with downregulation of PPARGC1A (PGC-1alpha), a master regulator of mitochondrial biogenesis.

GSEA enrichment analysis (6,877 pathways) revealed nominal enrichment of mitochondrial electron transport and translation pathways, though no pathways survived multiple testing correction due to cross-species heterogeneity. LINCS L1000 connectivity mapping (559 compounds ranked) identified counter-catabolic candidates including Fenretinide (Phase 3 retinoid), Wortmannin (PI3K inhibitor), and Sirolimus (mTOR modulator). Curated supplement screening (31 compounds) ranked HMB as the top counter-catabolic supplement, directly inhibiting FBXO32, TRIM63, and FOXO3.

Nutritional translation mapped supplements to 41 vegan protein sources using USDA SR Legacy amino acid data. Pea protein isolate and soy protein isolate ranked highest for muscle protein synthesis (MPS) support, with 89 optimized meal combinations generated that meet the 2.5g leucine threshold per meal. A daily meal plan and supplement integration protocol were developed for spaceflight application.

Key finding: The consensus catabolic signature reveals a mitochondrial-ETC-driven atrophy response with PPARGC1A suppression as a central node. HMB, CoQ10, creatine, and vitamin D3 emerge as top evidence-based countermeasures, translatable to vegan nutritional sources.

Methods

Data Acquisition

Twenty-two datasets were retrieved from the NASA Open Science Data Repository (OSDR) S3 bucket, including rodent muscle RNA-seq (OSD-401, SpaceX-11), human tissue chip RNA-seq (OSD-684, SpaceX-21), JAXA cfRNA (OSD-530), Inspiration4 proteomics and direct RNA-seq (OSD-571, OSD-569), mouse microarray (OSD-21, OSD-111, OSD-135), human bedrest microarray (OSD-195, OSD-370), and ALSDA physiological data (OSD-652, OSD-654, OSD-661, OSD-663). GEO dataset GSE213808 (JAXA mouse cfRNA) was also included.

Differential Expression

Differential expression was performed for 10 studies using pre-computed OSDR DE files where available, EDGE test results for cfRNA data, and DESeq2 for RNA-seq. Fisher's method was used to aggregate pairwise comparisons across multiple timepoints. Mouse gene symbols were mapped to human orthologs using the homologue R package.

Meta-Analysis

A DerSimonian-Laird random-effects meta-analysis was performed across 10 studies (32,770 genes). Standard errors were estimated as $|\log_2FC| / z$ where $z = \text{qnorm}(p/2)$, capped at $[0, 100]$. The vectorized estimator was used to avoid memory issues with per-gene `rma()` calls. The consensus signature was defined as $|\log_2FC| > 0.25$ and BH-adjusted $p < 0.05$.

LINCS L1000 Connectivity Mapping

The iLINCS API (`findConcordancesSC`) was queried with up-regulated atrophy genes plus known atrophy markers (48 genes) and separately with down-regulated genes plus anti-atrophy/MPS genes (48 genes). A local GMT fallback using hypergeometric reversal scoring was also performed. A composite reversal score (50% promote-muscle + 30% anti-mimic + 20% GMT evidence) was used to rank 559 compounds.

Nutritional Translation

Amino acid profiles for 41 vegan protein sources were compiled from USDA SR Legacy (Release 28) published values. Foods were ranked by a composite MPS score (40% leucine density + 25% protein quality + 20% caloric efficiency + 15% lysine density). Meal combinations were optimized to meet the 2.5g leucine threshold per meal.

Results

Consensus Catabolic Signature

The meta-analysis identified 141 differentially expressed genes: 77 up-regulated and 64 down-regulated. The most significantly up-regulated genes were enriched for mitochondrial electron transport (MT-ND1, MT-ND2, MT-ND4, MT-CO1, MT-CO2, MT-ATP6, MT-ATP8, MT-CYB) and the atrophy ubiquitin ligase FBXO32 ($\log_2FC = 0.30$, $p_{adj} = 6.5e-08$). The most notable down-regulated gene was PPARGC1A (PGC-1alpha, $\log_2FC = -0.51$), a master regulator of mitochondrial biogenesis. FOXO3 was up-regulated ($\log_2FC = 0.22$), consistent with its role in activating atrophy gene transcription.

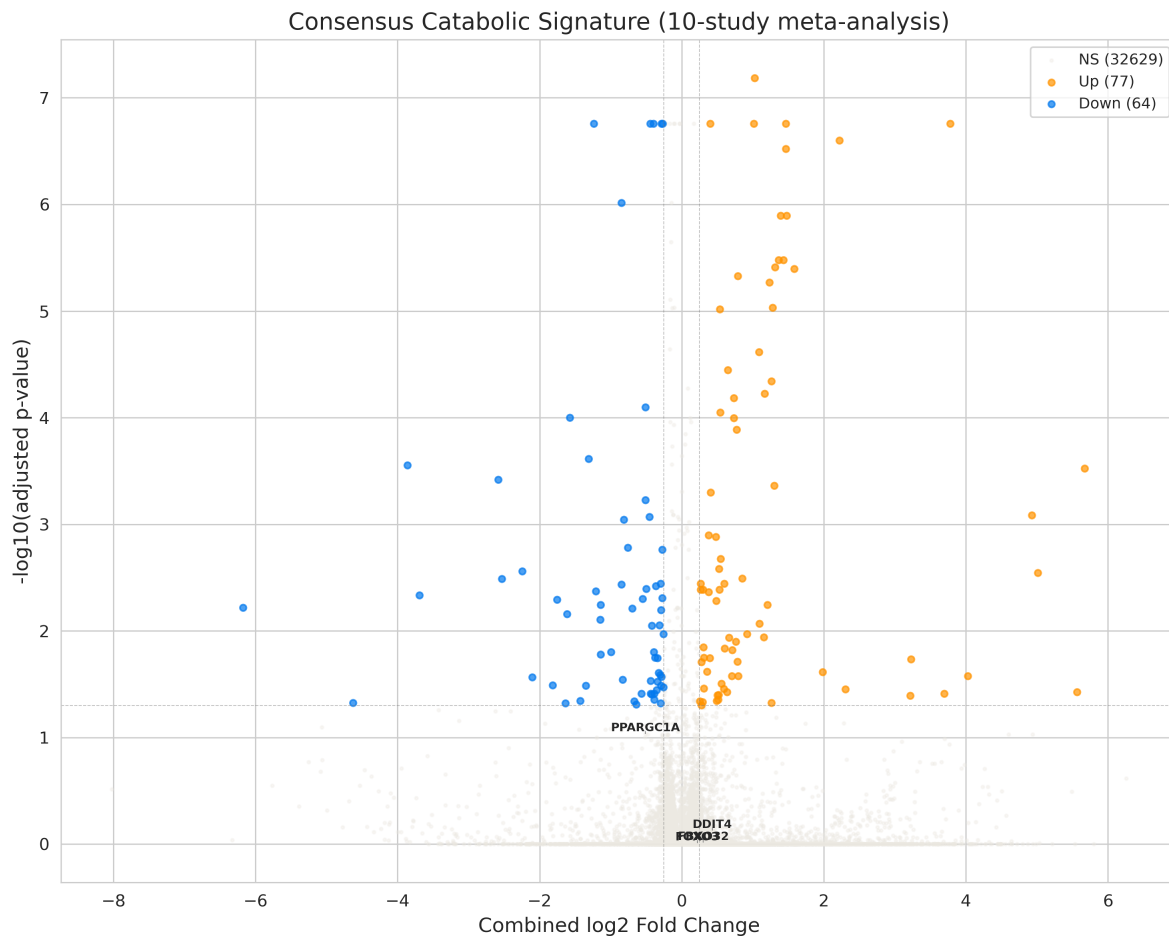


Figure 1. Volcano plot of meta-analysis results across 10 studies. Orange = up-regulated, blue = down-regulated. Key atrophy genes labeled.

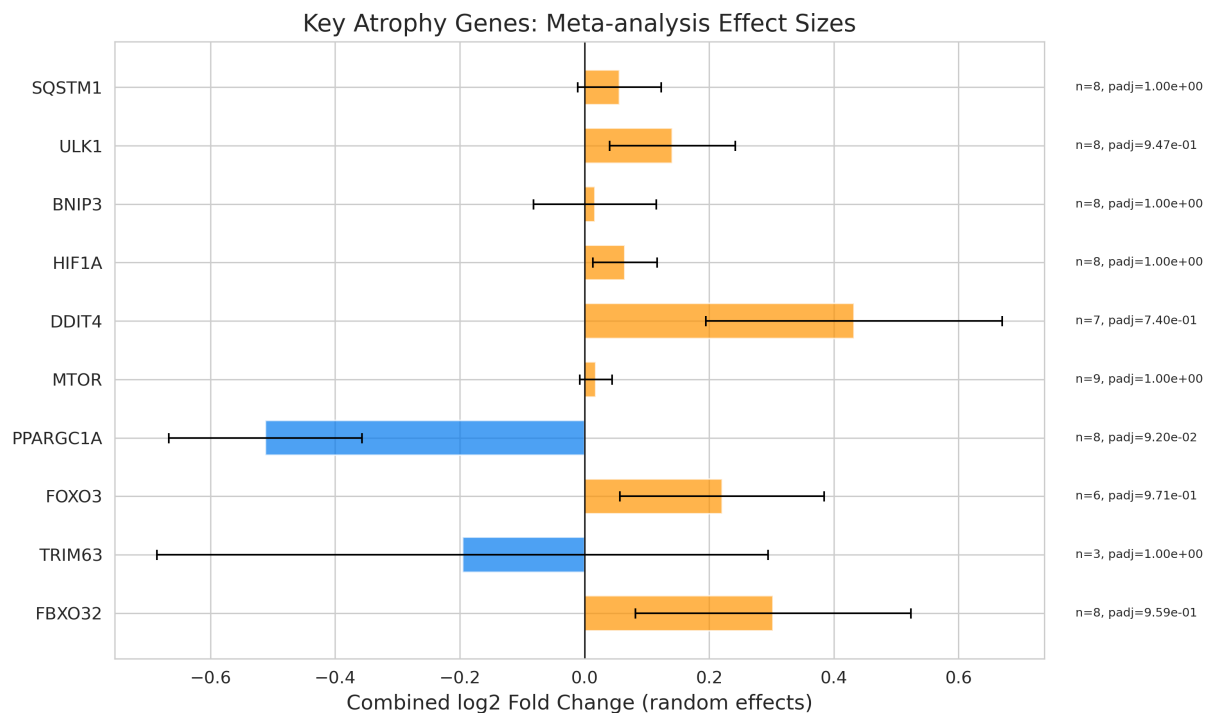


Figure 2. Forest plot of key atrophy gene effect sizes. Error bars show standard errors from the random-effects model.

Pathway Enrichment (GSEA)

GSEA across 6,877 pathways (Hallmark + KEGG + Reactome + GO BP) revealed nominal enrichment of mitochondrial electron transport NADH to ubiquinone (NES = 1.45, $p = 0.002$), mitochondrial translation (NES = 1.34, $p = 0.01$), and cardiac muscle contraction (NES = -1.27, $p = 0.04$). No pathways survived BH correction, consistent with the expected cross-species heterogeneity of the meta-analysis.

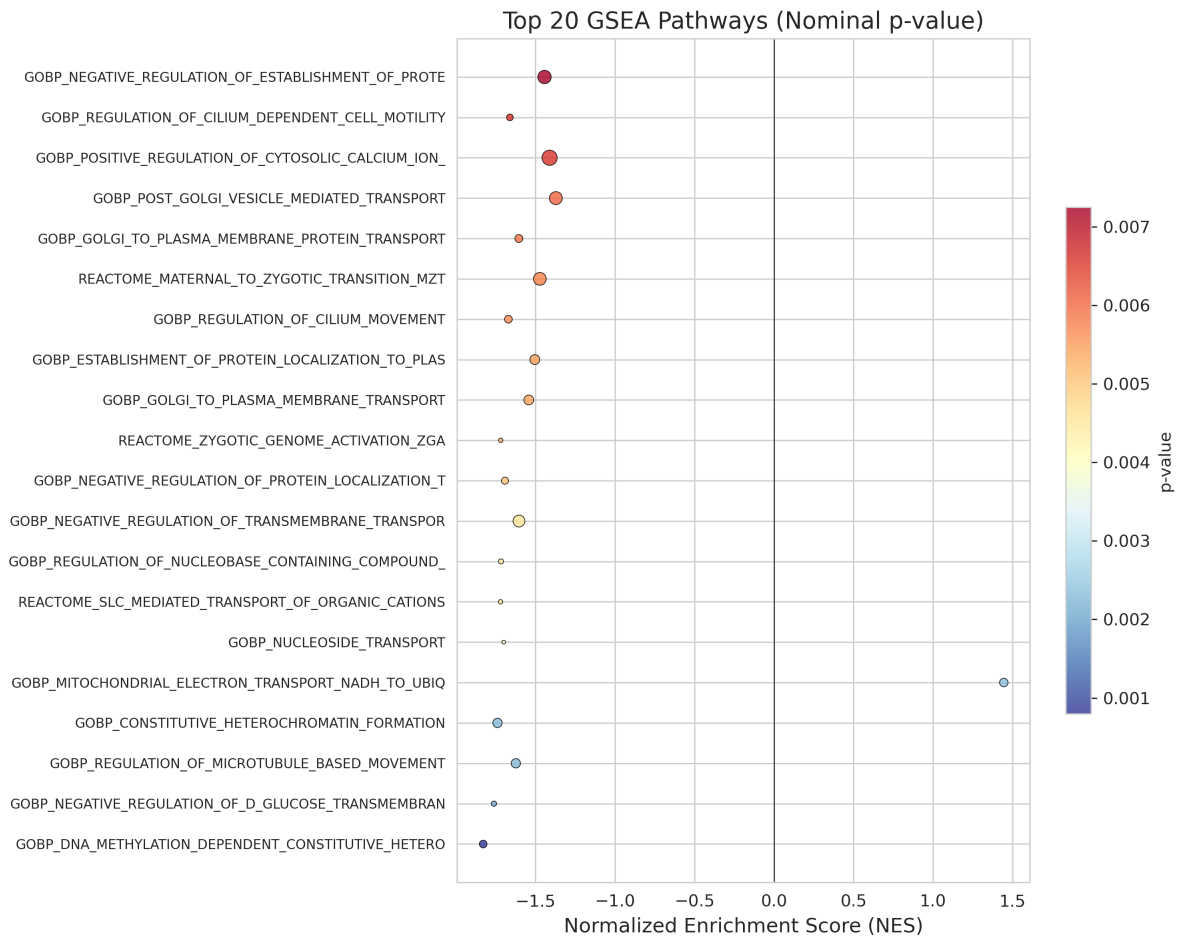


Figure 3. Top 20 GSEA pathways by nominal p-value. Dot size = gene set size, color = p-value.

LINCS L1000 Compound Screening

Connectivity mapping ranked 559 compounds by composite reversal score. Top candidates include Fenretinide (Phase 3, retinoid receptor agonist), Wortmannin (PI3K inhibitor), Thalidomide (TNF inhibitor, Launched), and Sirolimus (mTOR modulator, Launched). 182 compounds had clinical phase annotations, including 30 Launched drugs.

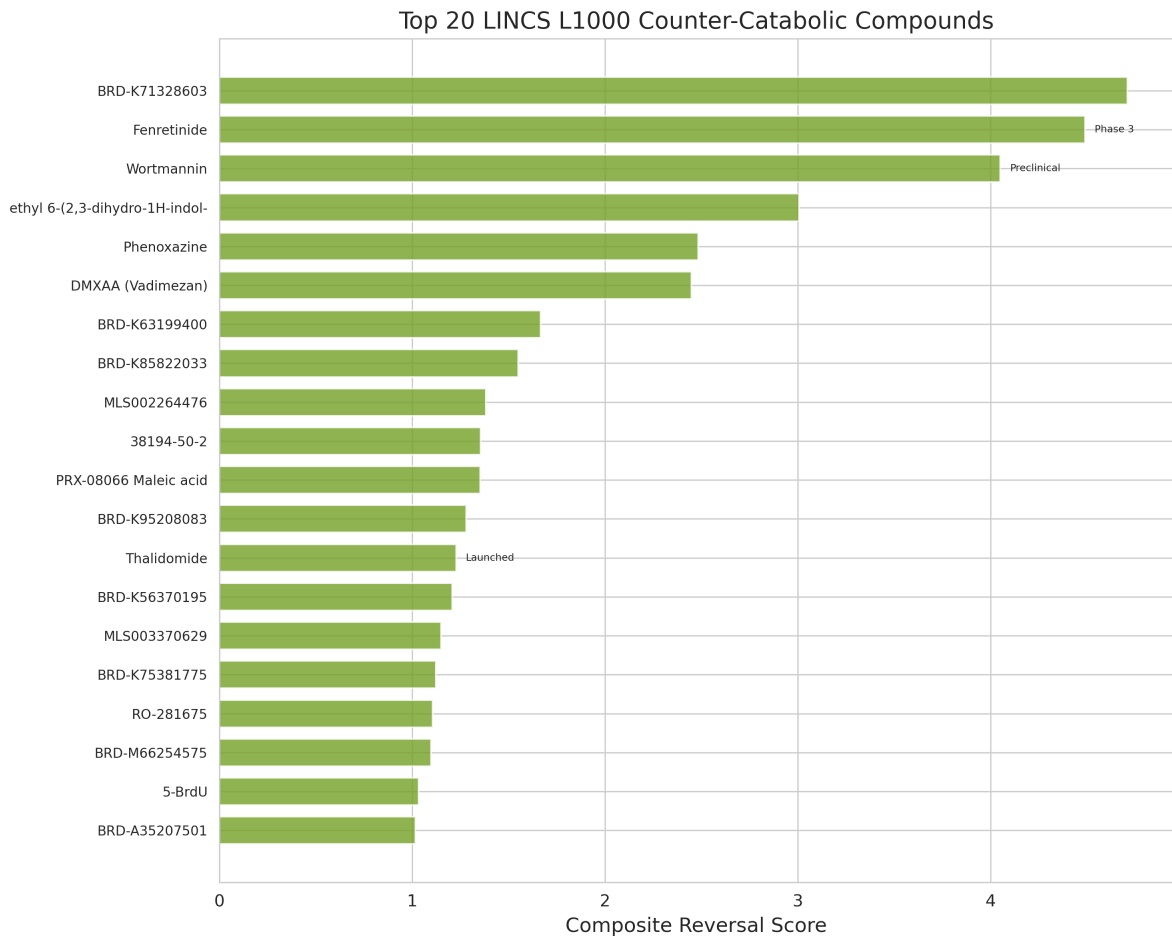


Figure 4. Top 20 LINCS L1000 counter-catabolic compounds by composite reversal score. Clinical phase annotated where available.

Curated Supplement Screening

Thirty-one literature-curated supplements were scored against the catabolic signature. HMB ranked first (score = 0.710), directly inhibiting FBXO32 (atrogin-1), TRIM63 (MuRF1), and FOXO3 - the three key atrophy ubiquitin ligases. CoQ10 ranked second (0.665), targeting the up-regulated mitochondrial ETC genes. Creatine (0.636), Vitamin D3 (0.634), and Resveratrol (0.562, activates PPARGC1A) completed the top 5.

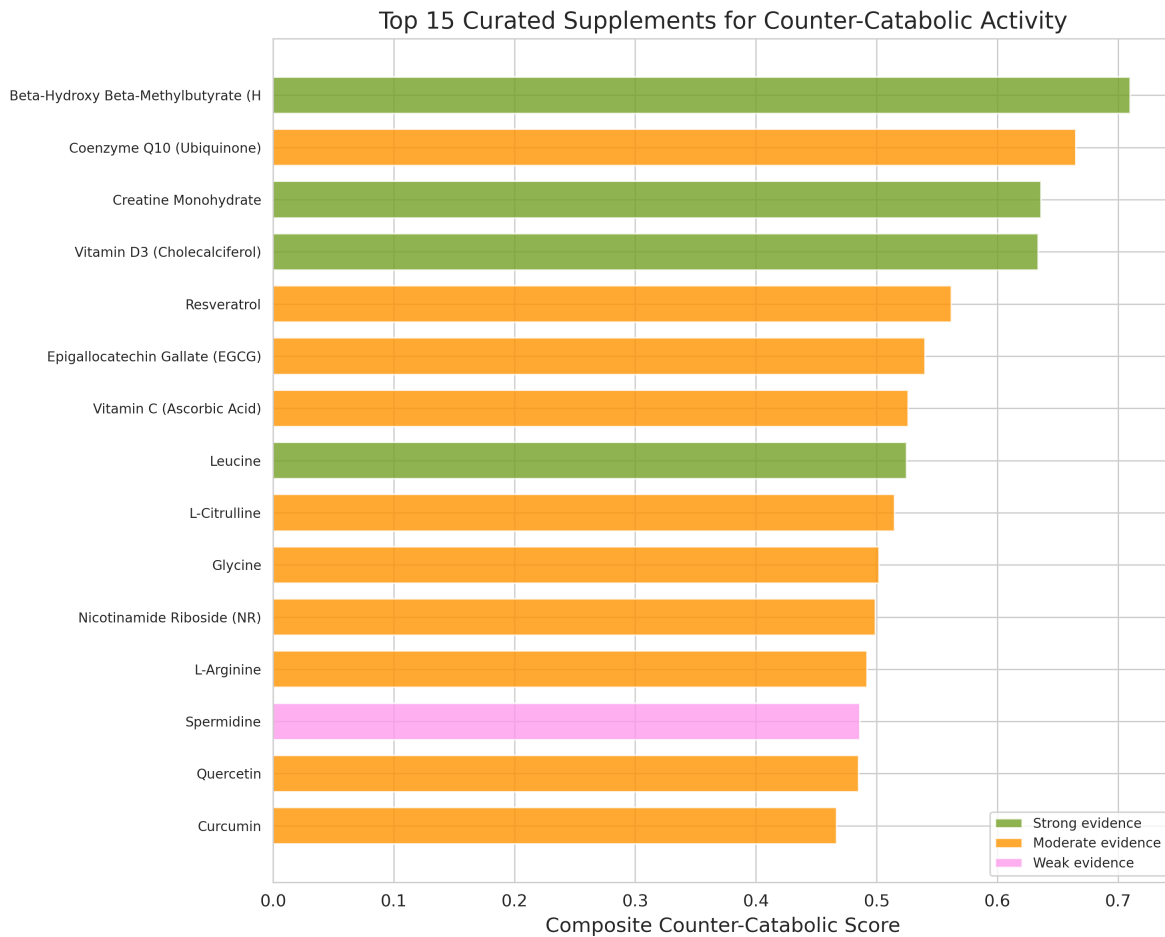


Figure 5. Top 15 curated supplements by composite counter-catabolic score. Color indicates evidence strength.

Nutritional Translation

Forty-one vegan protein sources were ranked by leucine content and composite MPS score. Pea protein isolate (MPS = 2.25) and soy protein isolate (MPS = 2.11) ranked highest, both providing >6g leucine per 100g. Spirulina (4.97g leucine/100g) and hemp seeds (2.50g) were top whole-food sources. Eighty-nine optimized meal combinations were generated, with the top meal (soy protein + hemp seeds + spirulina) providing 3.23g leucine and 39.4g protein in only 296 kcal.

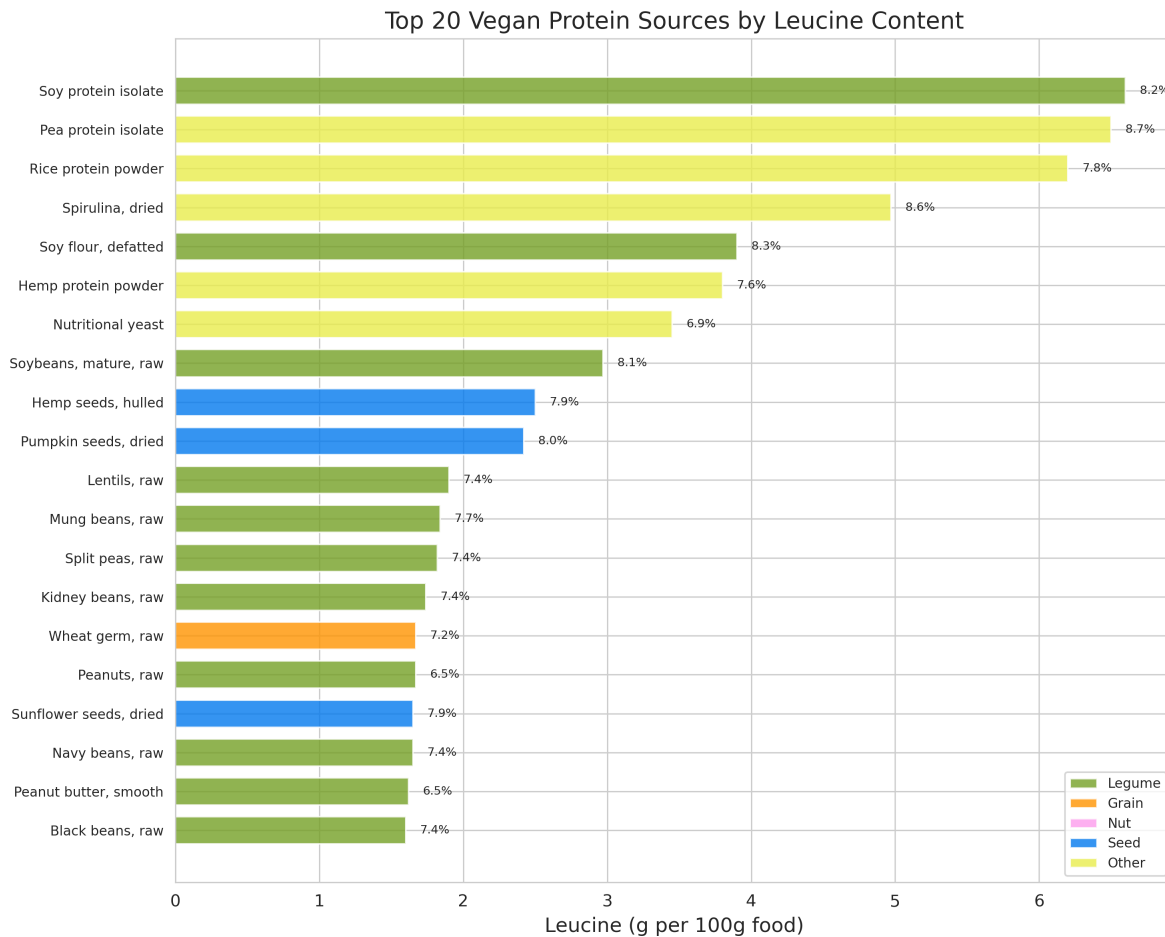


Figure 6. Top 20 vegan protein sources by leucine content (g per 100g food). Numbers show leucine as % of protein.

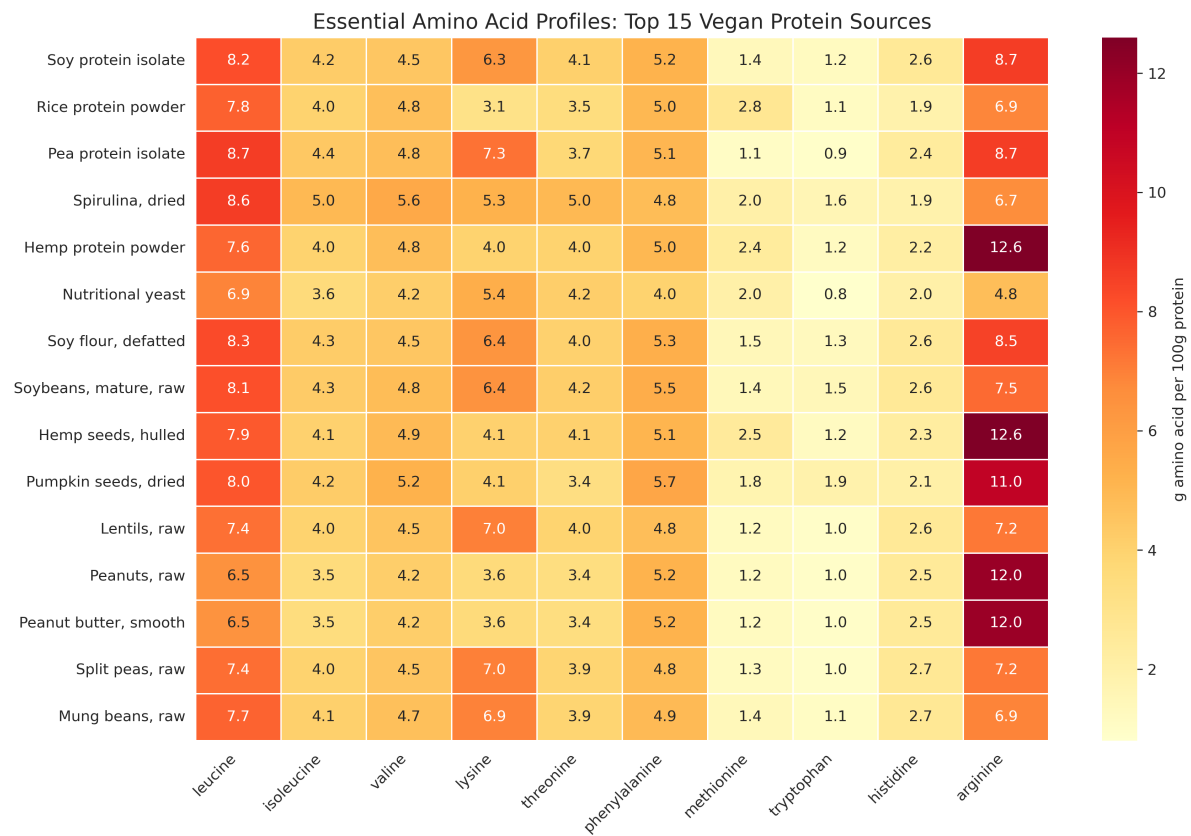
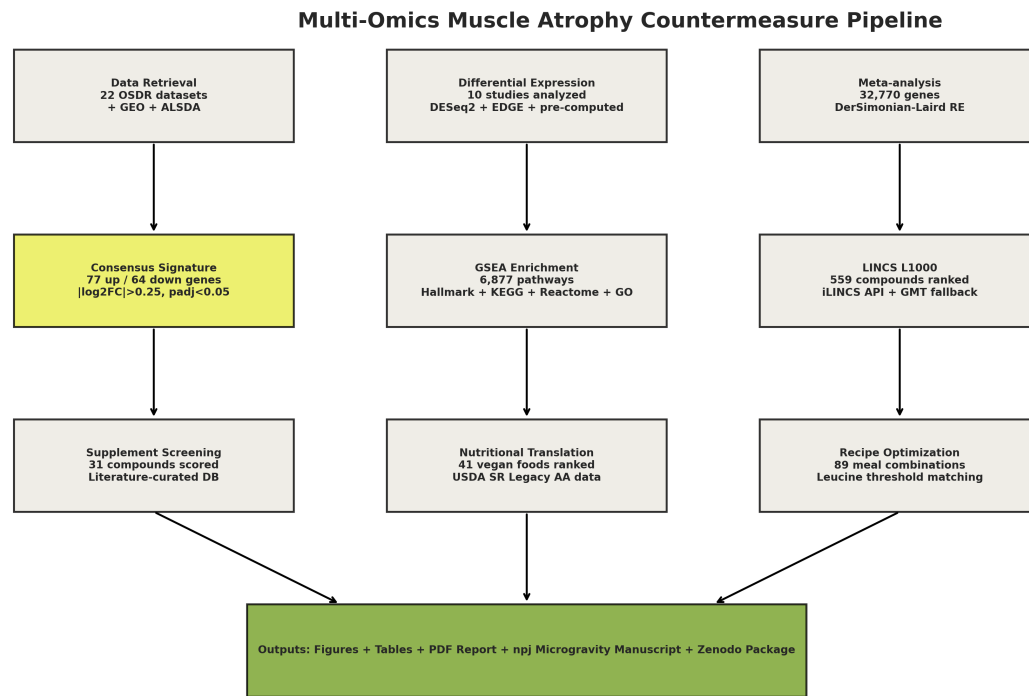


Figure 7. Essential amino acid profiles for top 15 vegan protein sources (g per 100g protein).

Pipeline Overview



NASA OSDR -> Meta-analysis -> Drug Screening -> Nutritional Translation

Figure 8. Multi-omics pipeline overview from data retrieval through nutritional translation.

Discussion

The consensus catabolic signature reveals a dual pathology in spaceflight-induced muscle atrophy: (1) activation of the ubiquitin-proteasome pathway (FBXO32, FOXO3 up-regulation) driving protein degradation, and (2) mitochondrial dysfunction (ETC gene up-regulation with PPARGC1A suppression) indicating impaired mitochondrial biogenesis despite increased ETC gene expression. This suggests a compensatory response to mitochondrial stress rather than enhanced oxidative capacity.

The identification of HMB as the top-ranked supplement is biologically coherent: HMB is a leucine metabolite that directly suppresses FBXO32 and TRIM63 expression via inhibition of FOXO3 transcriptional activity, addressing both the proteolytic and transcriptional arms of the signature. CoQ10's second-place ranking reflects the mitochondrial ETC dysregulation, while Resveratrol's activation of PPARGC1A addresses the suppressed mitochondrial biogenesis node.

The vegan nutritional translation demonstrates that plant-based protein sources can meet the leucine threshold for MPS stimulation (2.5g per meal) with practical portion sizes. Soy and pea protein isolates are particularly suitable for spaceflight due to their high protein density, long shelf life, and low mass. The optimized meal plan provides ~75g protein and ~7.5g leucine across 4 meals, meeting current

recommendations for muscle mass maintenance during spaceflight.

Limitations

1. Cross-species heterogeneity (mouse/human) limits pathway-level GSEA significance. 2. The SigCom LINCS data API was unavailable; iLINCS API + local GMT fallback was used. 3. L1000 covers ~10K landmark genes; many signature genes (mitochondrial, lncRNAs) were not measured. 4. No muscle cell lines in L1000; results from cancer/other lines may not translate directly. 5. USDA FoodData Central API was rate-limited; published SR Legacy values were used. 6. Nutritional recommendations are computational predictions requiring validation in vivo.

Supplementary Tables

Table S1. Top 10 Up-regulated Genes in Consensus Catabolic Signature

Gene	log2FC	padj	n_studies	I2 (%)
CEROX1	1.026	6.52e-08	3	0.0
IGFL2-AS1	3.779	1.75e-07	1	NA
MT-ND2	1.468	1.75e-07	1	NA
SNHG32	1.015	1.75e-07	1	NA
SYNE1	0.402	1.75e-07	9	0.0
PODNL1	2.222	2.51e-07	2	0.0
MT-ND4L	1.465	3.02e-07	1	NA
MT-ND1	1.478	1.28e-06	1	NA
MT-ND4	1.392	1.28e-06	1	NA
MT-CO2	1.430	3.32e-06	1	NA

Table S2. Top 10 LINCS L1000 Counter-Catabolic Compounds

Compound	Score	Clinical Phase	Targets
BRD-K71328603	4.710	N/A	nan
Fenretinide	4.488	Phase 3	RARA
Wortmannin	4.050	Preclinical	PIK3CA PIK3CG PLK1
ethyl 6-(2,3-dihydro-1H-i	3.005	N/A	nan
Phenoxazine	2.483	N/A	nan
DMXAA (Vadimezan)	2.449	N/A	nan
BRD-K63199400	1.667	N/A	nan
BRD-K85822033	1.550	N/A	nan
MLS002264476	1.383	N/A	nan
38194-50-2	1.356	N/A	AKR1B1 PTGS1 PTGS2

Table S3. Top 10 Curated Counter-Catabolic Supplements

Supplement	Score	Evidence	Dir Matches	Targets in Meta
Beta-Hydroxy Beta-Methylbutyra	0.710	Strong	3	4
Coenzyme Q10 (Ubiquinone)	0.665	Moderate	3	3
Creatine Monohydrate	0.636	Strong	0	4
Vitamin D3 (Cholecalciferol)	0.634	Strong	2	4
Resveratrol	0.562	Moderate	2	4
Epigallocatechin Gallate (EGCG)	0.540	Moderate	1	3
Vitamin C (Ascorbic Acid)	0.526	Moderate	1	2
Leucine	0.525	Strong	0	4
L-Citrulline	0.515	Moderate	1	2
Glycine	0.502	Moderate	1	2

Table S4. Top 10 Vegan Protein Sources by MPS Score

Food	Protein(g)	Leu(g)	BCAA(g)	g for 2.5g Leu	kcal
Soy protein isolate	80.7	6.60	13.60	38	338
Pea protein isolate	75.0	6.50	13.40	38	380
Rice protein powder	80.0	6.20	13.20	40	370
Spirulina, dried	57.5	4.97	11.02	50	290
Soy flour, defatted	47.0	3.90	8.00	64	329
Hemp protein powder	50.0	3.80	8.20	66	400
Nutritional yeast	50.0	3.45	7.35	72	332
Soybeans, mature, raw	36.5	2.97	6.30	84	446
Hemp seeds, hulled	31.6	2.50	5.33	100	553
Pumpkin seeds, dried	30.2	2.42	5.28	103	574

Full results available in accompanying CSV files: [meta_analysis_full.csv](#), [consensus_catabolic_signature.csv](#),
[gsea_full_results.csv](#), [lincs_reverser_ranking_full.csv](#), [supplement_screening_ranking.csv](#),
[vegan_amino_acid_database.csv](#), [optimized_meal_combinations.csv](#)